IT project management

Organizational issues

- Structure of classes
  - Exercises (min. 50 points) + Exam (min. 50 points)
  - Rating by scale
  - Written exam (2-3 open questions)
- Communication
  - Moodle website (general information, consultation dates and rules, MS Teams team access codes)
  - MS Teams team (announcements, materials, supporting literature)
  - E-mail (General and individual cases)

Points	Rating
180 - 200	5,0
161 – 180	4,5
141 – 160	4,0
126 – 140	3,5
100 – 125	3,0
0 – 100	2,0

General information about the subject

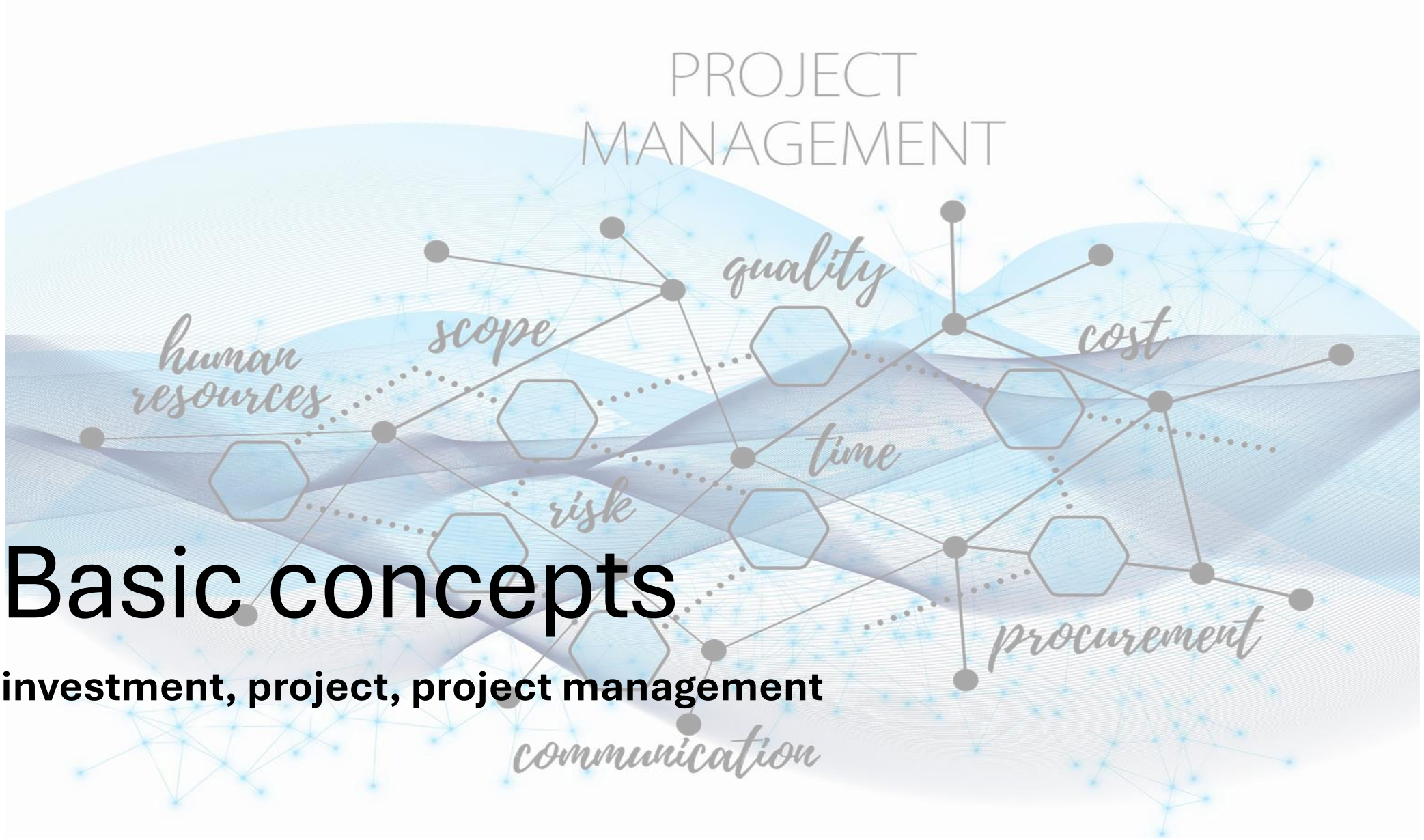
- GOAL

  Understand the essence of IT projects as a component of the implementation of organizational goals of both the investor (user, customer) and the contractor

- CLASS STRUCTURE
  - Basic concepts: investment, project, project management
  - Project management areas
  - Specifics of IT projects
  - Samples of IT project management methodologies

Literature (selected)

- E.Yourdon „Death March – The complete software developer’s guide to surviving „Mission Impossible” projects”
- F.P.Brooks, Jr. „Mystical Man-Month”
- T.DeMarco, T,Lister „ Peopleware – Productive Projects and Teams”
- H.Kerzner „Advanced Project Management”



Basic concepts

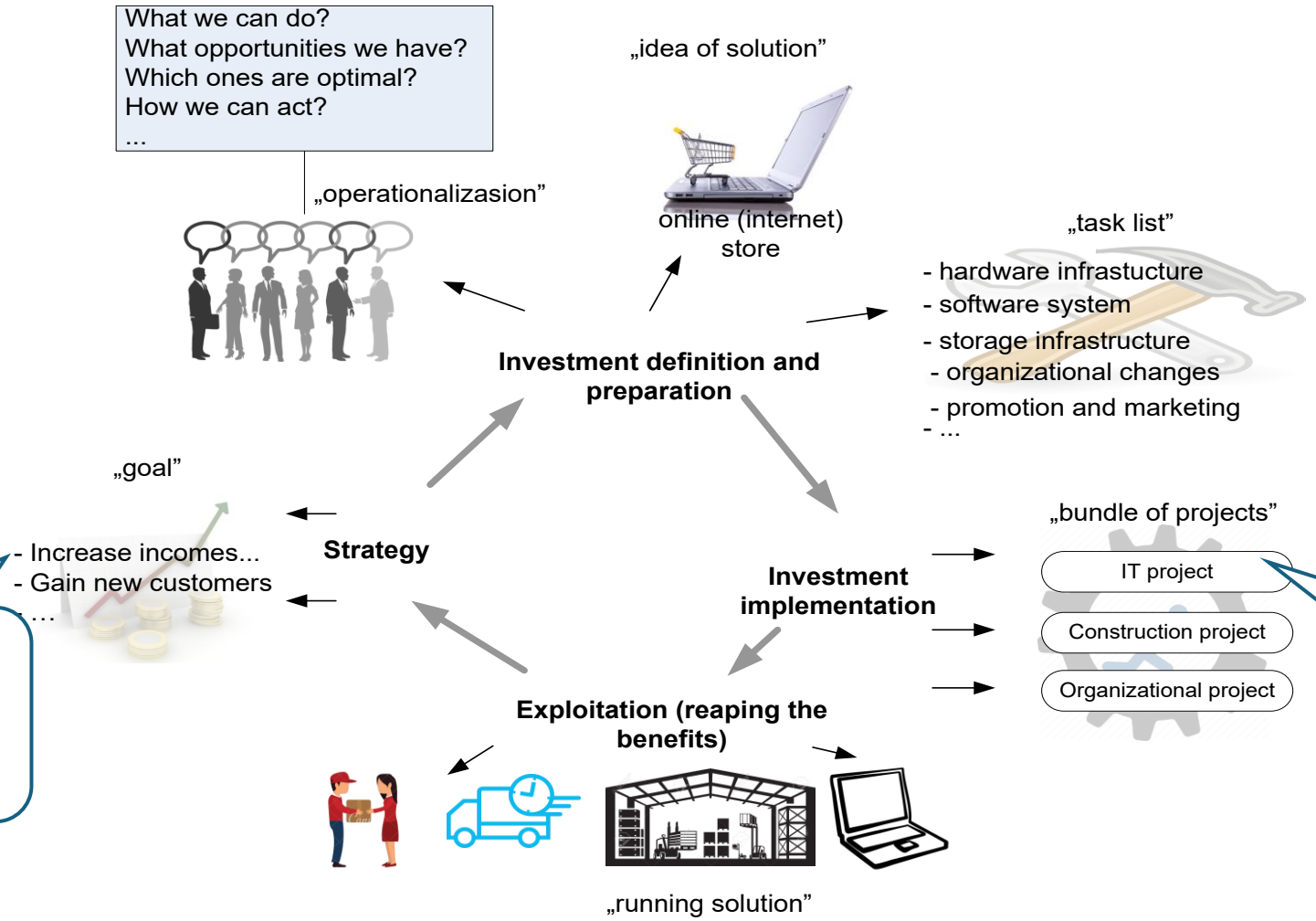
investment, project, project management

Genesis – effectiveness of IT project implementation

Result	2011	2012	2013	2014	2015	2017	2020
Succesful	29%	27%	31%	28%	29%	36%	31%
Challenged	49%	56%	50%	55%	52%	45%	50%
Failed	22%	17%	19%	17%	19%	19%	19%

- The so-called "chaos reports", published since 1994 by The Standish Group Inc., show the effectiveness of IT projects on a global scale based on three basic criteria: time, budget and scope
- After a significant improvement between the first and second reports in the 1990s, the situation stagnated, despite the emergence of the concept of agile software development (IT project management)
- These results have become a prerequisite for undertaking broader work on the adaptation of existing and the creation of new project management methods, taking into account the specificity of software development processes

Point of view: Investment or Project?



Investor's goal: Return on investment

Contractor's goal: Deliver software at a cost less than profit

From a formal point of view:

- INVESTMENT:
  - Giving up current and certain benefits in favor of future and uncertain benefits
  - Time (future benefits) – Expenses (Benefits waived) – Risks (uncertain benefits)
- PROJECT:
  - A sequence of unique, complex, and related activities that share a single goal that must be completed within a specific timeframe, within a specific budget and according to specification
  - scope (objective, specification) – deadlines (specific time) – resources (budget)

Points of view:

Although the points of view are different, understanding them is the key to working together for mutual success

Perception of success

Investor (investment):

- Achieving strategic goals within the adopted assumptions and tolerance limits

Contractor (project):

- Fulfillment of the contract (delivery of software) with the assumed profit margin

The project is a collective effort!

Where is the threat?

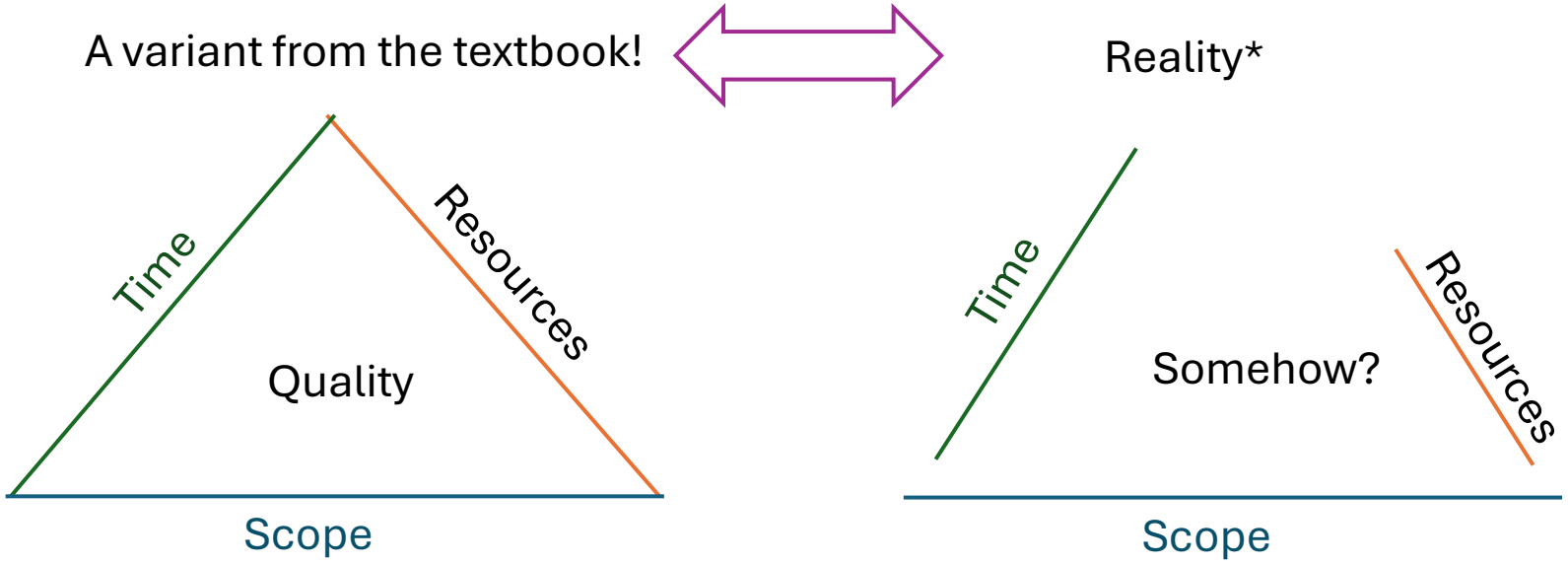
- Wrong assumptions about goals, ways and possibilities (time, costs) to achieve them

- Overestimation of feasibility (time, resources, ways of acting)

Investment risk

Project risk

Framework of action – project scope triangle



- Project management is the ability to deal with reality with the help of methods that can be learned from textbooks or discovered on your own based on experience**

*) see „Death March"
**) which usually takes longer and costs more

Project management areas

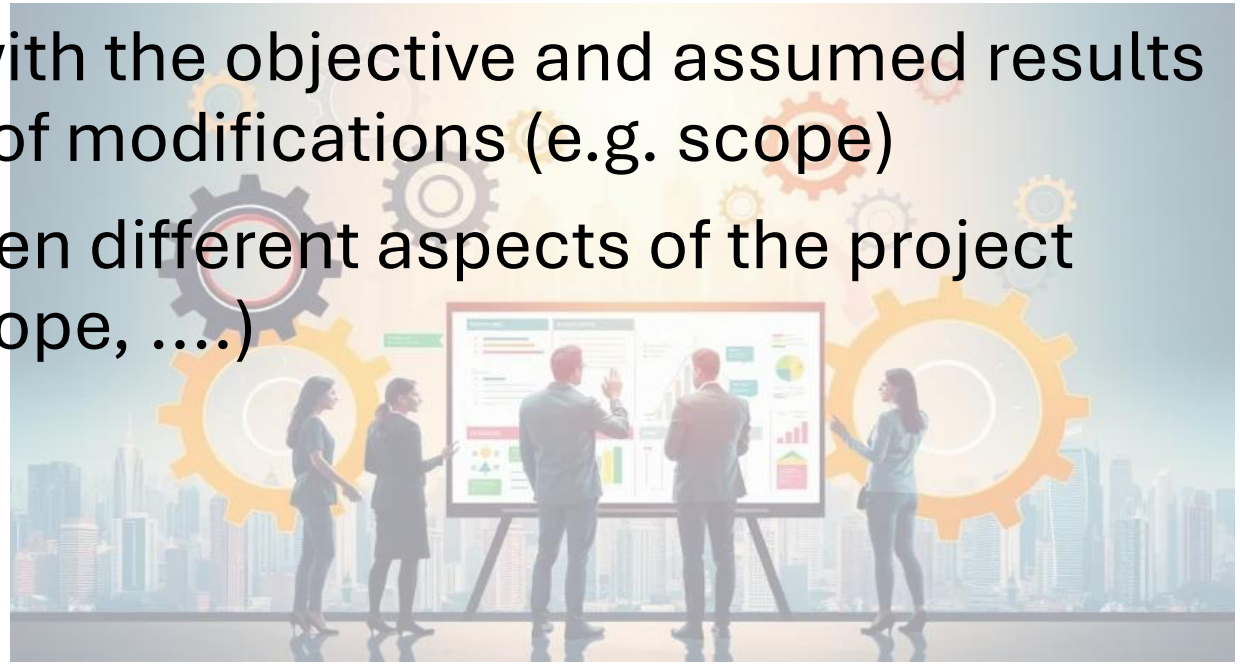
Selected aspects
Main ideas

PMBOK guide

- *„A guide to the project management body of knowledge”*
- This is not a theory!
- A collection of experience from many projects carried out in various fields, systematized and formalized
- It shows the multifaceted nature of the project environment and allows you to understand what a project is from different points of view
- It does not point out how to act, but indicates what should be done
  - The choice of methods of action is left open
- It is not a recipe for project management (methodology), but a starting point for creating a way of managing a project in a specific case!

PROJECT INTEGRATION MANAGEMENT

- Resource allocation
- Assignment of responsibilities and decision-making powers
- Verification of compliance with the objective and assumed results of the project in the context of modifications (e.g. scope)
- Ensuring consistency between different aspects of the project (time, budget, resources, scope, ...)
- Problems solving



PMBOK guide – chapter 4

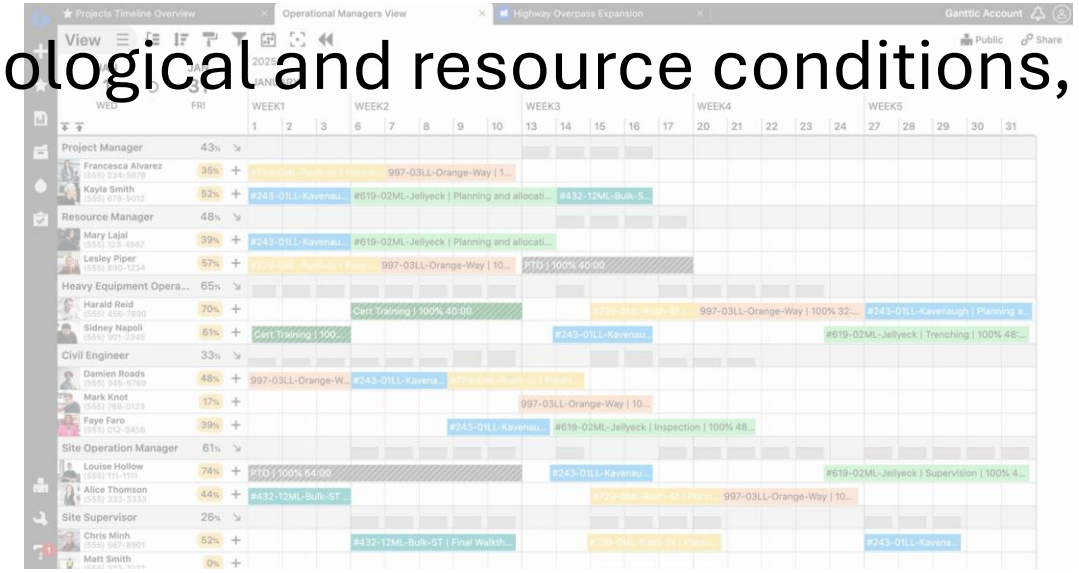
PROJECT SCOPE MANAGEMENT

- Identify and define the project scope
  - List of products and their specifications
- Manage project scope changes
  - Sources of change: incomplete/inconsistent specification, implementation problems, change of expectations, …
- Verification of project products (results)

PMBOK guide – chapter 5

PROJECT SCHEDULE MANAGEMENT

- Prepare, track and manage project schedule changes
- Identification of tasks, estimation of workload (time / resources / costs)
- Order of implementation (technological and resource conditions, priorities)



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PROJECT COST MANAGEMENT

- Create a project budget
  - Types of costs e.g.: Personnel costs, impersonal costs, material costs, external purchases, ...
- Cost management
  - Ongoing cost tracking, documenting financial flows, ...

PMBOK guide – chapter 7

PROJECT QUALITY MANAGEMENT

- Quality Management Plan
  - Selection of methods of quality control
  - Assignment of responsibilities and competencies
  - Define procedures
- Inspection of the implementation of the quality management plan
  - Verification and documentation of results

PMBOK guide – chapter 8

PROJECT RESOURCE MANAGEMENT

- Identify the necessary resources
  - People, tools (infrastructure), materials…
- Assign resources to tasks
- Building and managing teams
  - People management!

PMBOK guide – chapter 9

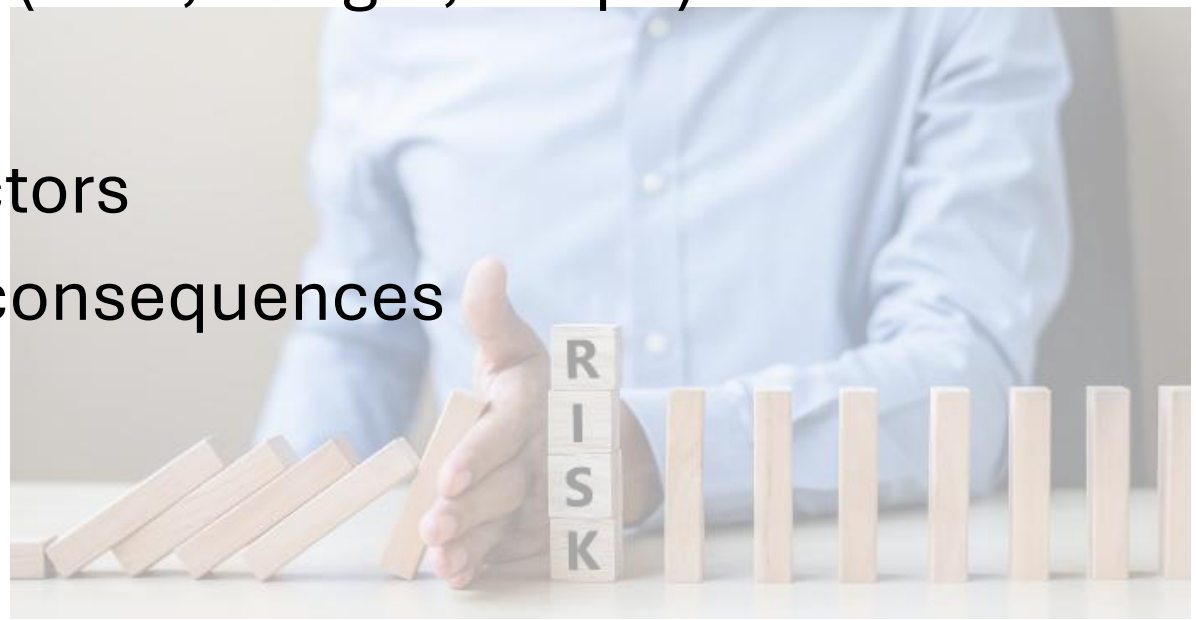
PROJECT COMMUNICATIONS MANAGEMENT

- Ensuring the proper flow of information in the project
  - External communication (stakeholders, environment,….)
  - Internal communication (information flow between project teams)
  - Team communication
- Defining communication channels
  - Who, to whom, what, when, in what form, ….

PMBOK guide – chapter 10

PROJECT RISK MANAGEMENT

- Any phenomenon or event that may have a negative impact on the project in terms of achieving the assumed goals or meeting the implementation assumptions (time, budget, scope)
- Identifying risk factors
- Monitoring the level of risk factors
- Preventing or minimizing the consequences



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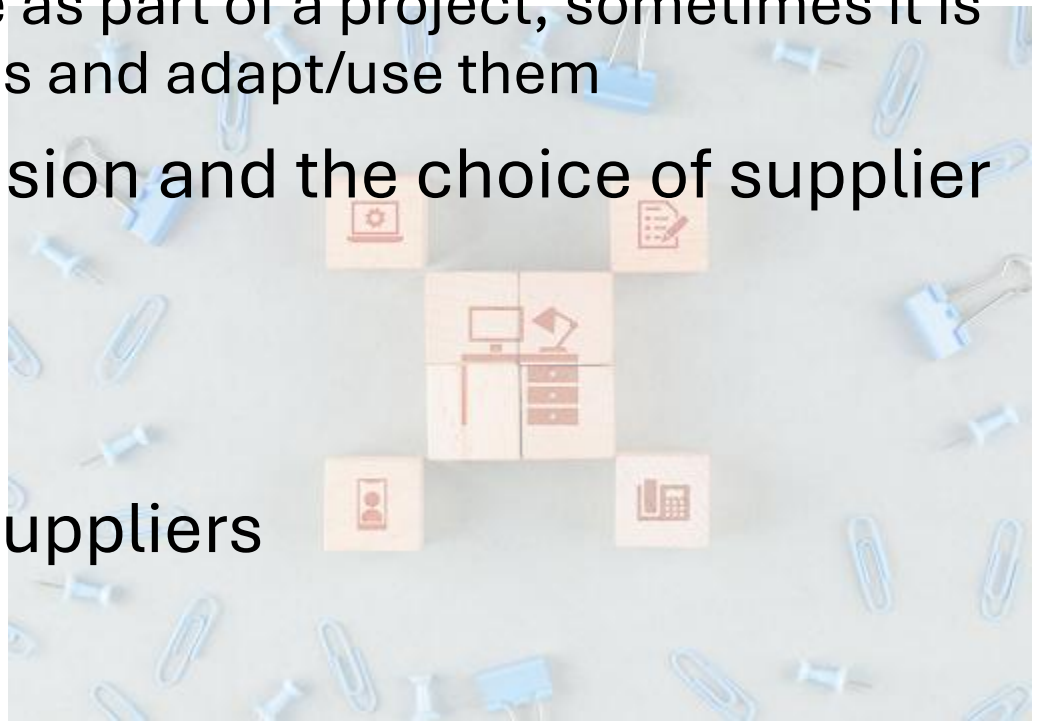
PROJECT PROCUREMENT MANAGEMENT

- External Supply Management
  - Not everything always has to be done as part of a project, sometimes it is better (?) to buy ready-made solutions and adapt/use them
- Justification of the purchasing decision and the choice of supplier
  - Market analysis (potential suppliers)
  - Analysis of available products
  - Cost analysis
- Acquisition and cooperation with suppliers



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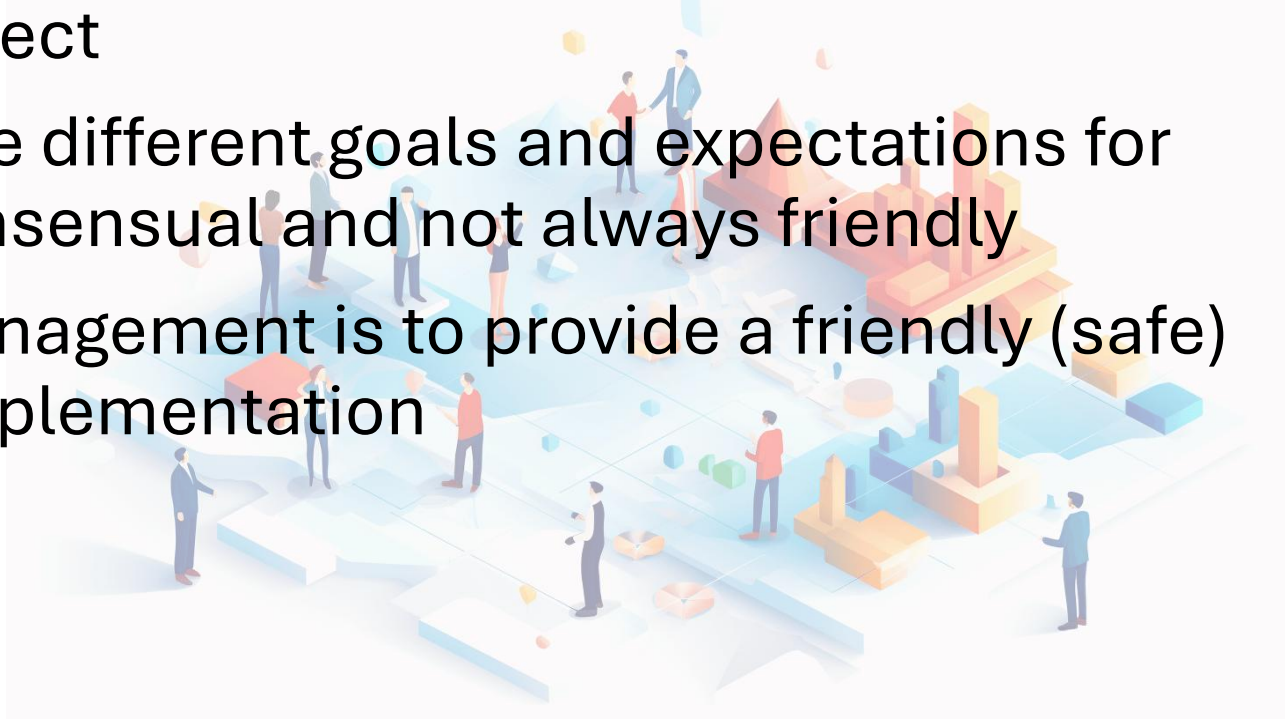
PROJECT STAKEHOLDER MANAGEMENT

- Any person or organization that may influence or is affected by the implementation of the project
- Each stakeholder may have different goals and expectations for the project, not always consensual and not always friendly
- The aim of stakeholder management is to provide a friendly (safe) environment for project implementation



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Project management areas

- Distinguished areas of management are present in every project, regardless of the so-called approach (agile, traditional, extreme…)
- The position we hold in the project determines the scope of perception of individual areas and the role we play in managing them
  - Project Manager
  - Team Leader
  - Team Member
  - ….
- If you want to develop your project management skills, you must, regardless of your position, be aware of the existence of management areas and be able to notice management activities